BKP: An R Package for Beta Kernel Process Modeling

Jiangyan Zhao

School of Statistics, ECNU

jyzhao@sfs.ecnu.edu.cn

joint work with Kunhai Qing and Jin Xu

2026.07.24

Contents

Background

- Estimating a **continuous probability function** from **binary** or **binomial observations** is fundamental in statistics and machine learning.
- Applications include:
  - Binary classification (MacKenzie et al., 2014; Wen et al., 2025)
  - Probability calibration (Sung et al., 2020; Dimitriadis et al., 2022)
  - Relative abundance modeling (Martin et al., 2020)
  - Longitudinal patient-reported outcomes (Najera-Zuloaga et al., 2018, 2019)
  - Infectious disease risk prediction (**?**)
  - ...

Classical methods & Challenges

- **Parametric method**: logistic regression is computationally efficient but limited in capturing complex nonlinear patterns.
- **Nonparametric method**: Gaussian Process (GP) offers flexibility and principled uncertainty quantification, but approximate inference for binary (*non-Gaussian*) likelihood increases computational cost.

Beta Kernel Process

- **Beta Kernel Process (BKP)**: a scalable and interpretable nonparametric model for binomial probability surfaces (Goetschalckx et al., 2011; MacKenzie et al., 2014; **?**)
- **Key features**:
  - Computational efficiency: closed-form posterior inference
  - More transparency and straightforward: beta-binomial conjugate pair
  - Uncertainty quantification: adaptive decision-marking
- **Dirichlet Kernel Process (DKP)**: a natural extension for categorical or multinomial data

Contents

Summary

- Developed the **first publicly available BKP package** implementing BKP and DKP models
- **Methodological contributions:**
  - Provided a **Bayesian interpretation** of the BKP model
  - Proposed a **flexible framework for constructing priors**, including data-adaptive options
  - Developed a **robust and efficient hyperparameter optimization strategy** using LOOCV and multi-start reparameterization.

Future Development Directions

- **Model extensions**: Extend the BKP framework to multivariate, functional, and spatio-temporal data, as well as mixed-type covariates
- **Alternative likelihoods**: Explore likelihoods beyond the binomial family, e.g., negative binomial for over-dispersed counts, geometric for waiting-time modeling
- **Community contributions**: We invite developers to contribute via GitHub pull requests `https://github.com/Jiangyan-Zhao/BKP/pulls`

Resources

- https://cran.r-project.org/web/packages/BKP/index.html
- https://github.com/Jiangyan-Zhao/BKP
- https://arxiv.org/abs/2508.10447


Q & A
THANK YOU

Dimitriadis, T., L. Dümbgen, A. Henzi, M. Puke, and J. Ziegel (2022). Honest calibration assessment for binary outcome predictions. *Biometrika 110*(3), 663–680.

Goetschalckx, R., P. Poupart, and J. Hoey (2011). Continuous correlated beta processes. In *Proceedings of the Twenty-Second International Joint Conference on Artificial Intelligence - Volume Two*, IJCAI'11, pp. 1269–1274. AAAI Press.

MacKenzie, C. A., T. B. Trafalis, and K. Barker (2014). A Bayesian beta kernel model for binary classification and online learning problems. *Statistical Analysis and Data Mining: The ASA Data Science Journal 7*(6), 434–449.

Martin, B. D., D. Witten, and A. D. Willis (2020). Modeling microbial abundances and dysbiosis with beta-binomial regression. *The Annals of Applied Statistics 14*(1), 94–115.

Najera-Zuloaga, J., D.-J. Lee, and I. Arostegui (2018). Comparison of beta-binomial regression model approaches to analyze health-related quality of life data. *Statistical Methods in Medical Research 27*(10), 2989—009.

Najera-Zuloaga, J., D.-J. Lee, and I. Arostegui (2019). A beta-binomial mixed-effects model approach for analysing longitudinal discrete and bounded outcomes. *Biometrical Journal 61*(3), 600–615.

Sung, C.-L., Y. Hung, W. Rittase, C. Zhu, and C. F. J. Wu (2020). Calibration for computer experiments with binary responses and application to cell adhesion study. *Journal of the American Statistical Association 115*(532), 1664–1674.

Wen, H., A. Betken, and H. Hang (2025). Optimal learning of kernel logistic regression for complex classification scenarios. In *The Thirteenth International Conference on Learning Representations*.